

Discussion 8

Note: Your TA will probably not cover all the problems on this worksheet. The discussion worksheets are not designed to be finished within an hour. They are deliberately made slightly longer so they can serve as resources you can use to practice, reinforce, and build upon concepts discussed in lectures, discussions, and homework.

This Week's Cool AI Demo/Video:

Anduril EagleEye: <https://www.youtube.com/watch?v=x9B02pFKpJo>

Guess who (temporarily) tanked googles market cap by 100B with one tweet: https://www.youtube.com/watch?v=Ej6hnsQgV_c

1 Backprop in Practice: Staged Computation

We will apply backpropagation for the function $f(x, y, z) = (x + y)z$.

(a) Decompose f into two simpler functions.

(b) Draw the computation graph that represents f following the composition from part (a).

- (c) Write the steps of the forward pass and backward pass through the network.
- (d) Update your network drawing with the intermediate values in the forward and backward pass using the inputs $x = -2$, $y = 5$, and $z = -4$.

2 A Two Layer Neural Network

Consider a neural network with two fully connected layers **but no nonlinearities**. The network maps input $\mathbf{x} \in \mathbb{R}^2$ to output $y \in \mathbb{R}$ according to the following functions:

$$\mathbf{h} = \mathbf{W}_1 \mathbf{x}, \quad y = \mathbf{W}_2 \mathbf{h},$$

where $\mathbf{W}_1 \in \mathbb{R}^{2 \times 2}$ and $\mathbf{W}_2 \in \mathbb{R}^{1 \times 2}$. We define the loss as the mean squared error:

$$L = \frac{1}{2}(y - t)^2,$$

where t is the target output.

- (a) Write the steps of the forward pass and backward pass through the network *Hint*: You may find the following gradient identities useful:

$$(i) \frac{\partial}{\partial \mathbf{x}}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}, \quad (ii) \frac{\partial}{\partial \mathbf{A}}(\mathbf{a}^T \mathbf{A} \mathbf{x}) = \mathbf{a} \mathbf{x}^T, \quad (iii) \frac{\partial}{\partial \mathbf{x}}(\mathbf{A} \mathbf{x}) = \mathbf{A}.$$

$$(iv) \frac{\partial L}{\partial \mathbf{x}} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial \mathbf{x}}, \quad (v) \frac{\partial L}{\partial \mathbf{x}} = \left(\frac{\partial y}{\partial \mathbf{x}} \right)^T \frac{\partial L}{\partial y}, \quad (vi) \frac{\partial L}{\partial \mathbf{A}} = \sum_{i,j} \frac{\partial L}{\partial y_{ij}} \frac{\partial y_{ij}}{\partial \mathbf{A}}$$

where y is a scalar, $\mathbf{a}$, $\mathbf{x}$ and $\mathbf{y}$ are vectors and $\mathbf{A}$ and $\mathbf{Y}$ are matrices.

- (b) Write out y directly in terms of $\mathbf{x}$ and the weights $\mathbf{W}_1, \mathbf{W}_2$.
- (c) Suppose you insert a nonlinearity $h(\cdot)$ after the first layer so that $\mathbf{h} = h(\mathbf{W}_1 \mathbf{x})$. Why does this change what the network can represent? Illustrate using a simple example with ReLU.

3 Model Intuition

- (a) What can go wrong if you just initialize all the weights in a neural network to exactly zero? What about to the same nonzero value?

- (b) Adding nodes in the hidden layer gives the neural network more approximation ability, because you are adding more parameters. How many weight parameters are there in a neural network with architecture specified by $n = [n^{(0)}, n^{(1)}, \dots, n^{(\ell)}]$, a vector giving the number of nodes in each of the $\ell + 1$ layers? (Layer 0 is the input layer, and layer ℓ is the output layer.) Evaluate your formula for a network $n = [8, 10, 10, 3]$.

- (c) Consider the two networks in the image below, where the added layer in Network B has 10 units with **linear activation**. Give one advantage of Network A over Network B, and one advantage of Network B over Network A.

